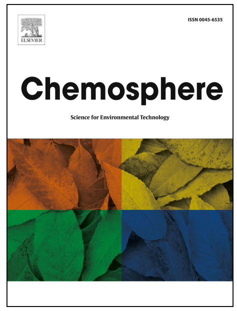

Antioxidant responses of triangle sail mussel *Hyriopsis cumingii* exposed to harmful algae *Microcystis aeruginosa* and high pH

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Antioxidant responses of triangle sail mussel *Hyriopsis cumingii* exposed to harmful algae *Microcystis aeruginosa* and high pH

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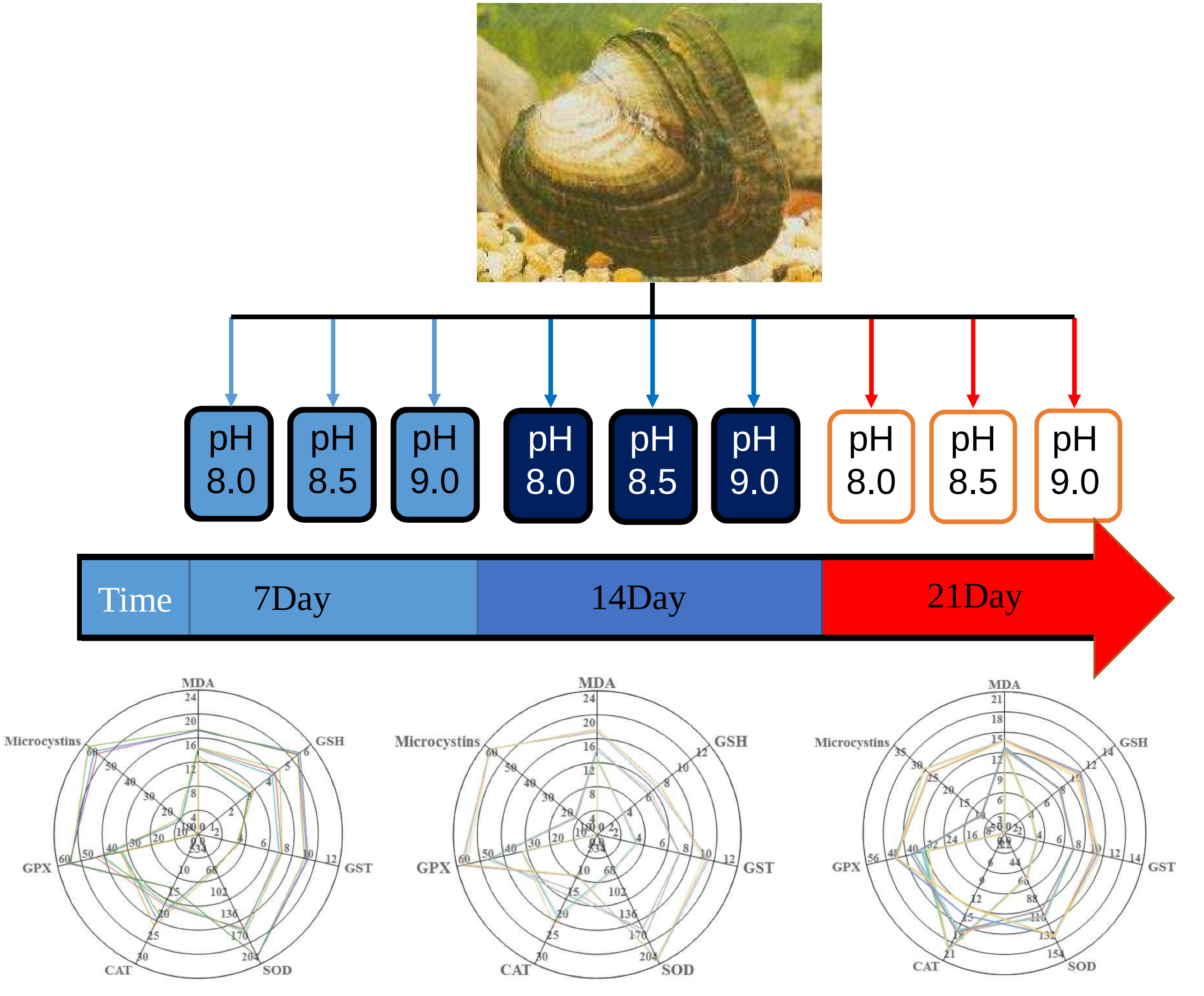

pH 8.0
pH 8.5
pH 9.0
pH 8.0
pH 8.5
pH 9.0
pH 8.0
pH 8.5
pH 9.0
Time
7Day
14Day
21Day
MDA
GSH
GST
SOD
CAT
GPX
Microcystins

Antioxidant responses of triangle sail mussel *Hyriopsis cumingii* exposed to harmful algae *Microcystis aeruginosa* and high pH

ABSTRACT In lakes and reservoirs, harmful algal blooms and high pH have been deemed to be two important stressors related to eutrophication, especially in the case of CO_2 depletion caused by dense blooms. However, the effects of these stressors on the economically important shellfish that inhabit these waters are still not well-understood. This study evaluated the combined effects of the harmful algae *Microcystis aeruginosa* (0%, 50%, and 100% of total dietary dry weight) and high pH (8.0, 8.5 and 9.0) on the antioxidant responses of the triangle sail mussel *H. cumingii*. The mussels were exposed to algae and high pH for 14 d, followed by a 7-day depuration period. Reactive oxygen species (ROS) in the mussel hemolymph, antioxidant and detoxifying enzymes, such as glutathione-S-transferase (GST), glutathione (GSH), superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), and malondialdehyde (MDA) in the digestive glands were analyzed during the experimental period. GST, SOD and GPx activity levels and the content of GSH increased following exposure to toxic *M. aeruginosa*, whereas CAT activity was inhibited. pH showed no significant effects on the immune defense mechanisms and detoxification processes. However, a high pH could cause increased ROS and MDA levels, resulting in oxidative injury. After a 7-day depuration period, exposure to toxic *M. aeruginosa* or high pH resulted in latent effects for most of the examined parameters. The treatment group exposed to the highest pH (9.0) displayed an increased oxidation state compared with the other pH treatments (8.0 and 8.5) for the same concentrations of toxic *M. aeruginosa*. The trends observed for ROS, MDA, GPx, GST, SOD and GSH levels indicated that a high density of toxic algae could result in severe and

continuous effects on mussel health.

Keywords: *Microcystis aeruginosa*, High pH, Triangle sail mussel *Hyriopsis cumingii*, Antioxidant capacity

1. Introduction

Eutrophication caused by the enhancement of anthropogenic activities has resulted in the excessive growth and geographical distribution of harmful algal blooms (HAB) worldwide (Ibelings et al., 2007; Metcalfe et al., 1998). Eutrophication can have serious impacts, including the disappearance of submerged macrophytes and increased phytoplankton levels in aquatic ecosystems (Körner, 2001). During the past few years, harmful algal blooms have been observed with increasing frequency in many regions (Aguiar et al., 2011; Karim et al., 2002; Kim et al., 2017). Harmful *Microsystis aeruginosa*, one of the most common bloom forming cyanobacteria species (Yang and Wang, 2019), is a noxious species that releases several types of microcystin into eutrophic waters (Sabatini et al., 2011). Microcystins could exert toxicity by inhibiting serine and threonine protein phosphatases, inducing oxidative stress in hepatocytes, and influencing the physiological processes of differentiation, cell growth and intracellular signaling and adversely affecting aquatic organisms in the waters (Amado and Monserrat, 2010; Kim et al., 2017; Yang et al., 2018).

Moreover, harmful algal blooms are often accompanied by a pH shift. Dense, harmful algal blooms generally consume the dissolved CO_2 in waters, leading to increases in pH values. In some extreme cases, the CO_2 levels can drop to 0.1 μM, which is equal to a pCO_2 value of less than 3 parts per million (ppm) (Balmer and Downing, 2011; Lazzarino et al., 2009). The exhaustion of CO_2 by dense blooms can lead to high pH values, with values of as high as 9 (Ibelings and

Maberly, 1998; Talling, 1976; Verspagen et al., 2014) and 11 being reported for some hypereutrophic shallow lakes (Lopez-Archilla et al., 2004). pH is an important parameter for water and can be a key driver of biological responses. Numerous studies have shown that pH shifts can cause a variety of negative effects in aquatic organisms. Increasing pH values is a common phenomenon associated with harmful algal blooms (Scott et al., 2005). However, most recent studies have focused only on the effects of harmful algal blooms on aquatic animals, neglecting to examine the effects caused by elevated pH levels. In the current study, the combined effects of harmful algal blooms and high pH values on aquatic animals were evaluated.

Bivalves are economically important aquaculture species all over the world. As filter-feeding organisms, bivalves can filter large volumes of water and can actively filter and trap suspended nutrients and food particles (Zhang et al., 2016). Subsequently, they have been shown to be good indicators of numerous environmental changes and/or anthropogenic factors due to their sensitive physiological responses to execrable water-soluble metabolites and bioaccumulated compounds (Pan et al., 2006; Solé et al., 1995; Vareli et al., 2012). When bivalves are exposed to environmental stressors, the organisms experience oxidative stress as the result of over-producing reactive oxygen species (ROS), and subsequently, antioxidant responses may be triggered (Pan et al., 2006). Antioxidant enzymes, such as superoxide dismutase (SOD), catalase (CAT) and glutathione peroxidase (GPx), act by detoxifying excessive ROS. However, these antioxidant enzymes may not be capable of removing over-produced ROS, which can cause cellular damage, such as lipid peroxidation (LPO) (Lushchak, 2011). Glutathione S-transferase (GST) enzymes participate in the cellular detoxification process by removing harmful chemical compounds (Alves de Almeida et al., 2007; Salinas and Wong, 1999). Therefore, detecting the activity levels of antioxidant or defense systems within bivalves could facilitate the evaluation of the medium-term and sub-lethal effects caused by harmful algae and environmental stressors.

The primary objective of this study was to investigate the effects of the harmful algae *M. aeruginosa* and high pH on the antioxidant defense system of the triangle sail mussel *Hyriopsis cumingii*. *H. cumingii* is the major species of mussel used for freshwater pearl production in China and also an important environmental indicator organism. (Fei et al., 2005; Zhang et al., 2007; Hu et al., 2013). We hypothesized that a high pH level would aggravate the antioxidant responses of mussels exposed to toxic *M. aeruginosa*, with some combined effects, and that the triangle sail mussel *H. cumingii* would be able to recover promptly from the combined oxidative effects caused by toxic *M. aeruginosa* and high pH levels. Our results will help to better understand the ecological toxicological risks of harmful algae blooms to the bivalves.

2. Materials and methods

2.1. Experimental mussels

H. cumingii triangle sail mussels (85.18±5.37 mm shell length, 82.58±6.12 g wet weight with shell), irrespective of sex, were collected during the summer of 2016 from the Jinhua Weiwang Aquaculture Farm, Zhejiang Province, China. The mussels were washed to remove fouling organisms and debris, and five-hundred and forty mussels live mussels were brought to a fiber-glass tank (500 L) with a filtrating equipment, acclimatized at 28±1°C, pH 8.0±0.1, in a 12 h light: 12 h dark cycle fed with 25 g *C. vulgaris* daily (around 1% of their dry tissue weights, above the maintenance requirements of the animals) before conducting the experiment for 14 days.

2.2. Algal cultures

The green microalga *C. vulgaris* (clone FACHB-8) and the toxic alga *M. aeruginosa* (clone

FACHB-905) were purchased from the Freshwater Algae Culture Collection of the Institute of Hydrobiology, Chinese Academy of Sciences (Wuhan, China). *C. vulgaris* were cultured in Watanabe medium (Watanabe, 1960), and *M. aeruginosa* were cultured in BG11 medium. The temperature was set to 28 °C, with a 12 h light: 12 h dark cycle. Microalgae were harvested during the stationary growth phase. Cell densities were counted using a hemocytometer under a light microscope.

2.3. Experimental setup

Acclimation was followed by 14 days of exposure to nine different treatments (3 × 3 factorial design), using three levels of pH (8.0, 8.5 and 9.0) and three concentrations of toxic *M. aeruginosa* (described below), with three replicates (i.e., 3 tanks) per treatment. Following this, a depuration period of seven days was applied during which mussels were fed only *C. vulgaris* at 50 ± 1.85 mg L^{-1} and under pH 8.0.

pH levels were manipulated by adding different proportions of $NaHCO_3$ and Na_2CO_3. pH values were checked approximately every 4 h with the pH-meter (Mettler-Toledo, Greifensee, Zurich). And adding appropriate volumes of 0.1 M NaOH to stabilize the pH value. The experimental diets, which particulate organic matters were adjusted to approximately 50 mg L^{-1} (50.04 ± 1.58 mg L^{-1}, dry weight) were prepared. The ratios of cell biomass of toxic *M. aeruginosa* in the three diet treatments were 0%, 50% and 100% (concentration in the water: 0, 8.6×10^6, and 1.7×10^7 cell mL^{-1}). Non-toxic *C. vulgaris* was used as a supplement to maintain the same algal biomass (concentration: 2.4×10^6, 1.23×10^6, and 0 cell mL^{-1}) in the water among the nine treatments, thereby avoiding the potential influence of food abundance on the physiological performance of mussels. Full water renewal and microalgae additions were conducted twice a day (at 08:00 and

20:00). Five mussels from each tank were collected on day 7 (the middle of the exposure period), on day 14 (the end of the exposure period) and on day 21 (the end of the depuration period). For each sampling time, three mussels were used for immunological measurements and two were used for toxin extraction and analysis.

2.4. Sample preparation

For each replicated aquarium, samples from three mussels of each replicate were pooled to eliminate individual differences and to obtain enough tissue for the assays. Hemolymph (2 mL per mussel) was extracted from the adductor muscle sinus, using a 2-mL sterilized syringe with a needle (0.75×38 mm). Subsequently, the hemolymph pools were transferred to 15 mL centrifuge tubes and placed on ice until ROS level analysis. After the collection of the hemolymph, the digestive glands were excised from the three mussels, thoroughly washed in 50 mM phosphate buffer (pH 7.4), and placed on ice. Samples were snap frozen in liquid nitrogen and stored at -80°C for further enzymatic assays. Digestive glands were later defrosted on ice and homogenized at a ratio of 1:4 (w/v) in 0.1 M Tris-HCl buffer, pH 7.5, containing 0.15 M KCl, 0.5 M sucrose, 1 mM EDTA, 1 mM dithiothreitol (DTT, Sigma) and 40 $\mu g\,mL^{-1}$ Aprotinin (Sigma), using 12–15 strokes of a motor driven Teflon Potter-Elvehjem homogenizer. Homogenates were then sonicated for 2 min at 0°C and were centrifuged at 10,000×g for 25 min at 4°C. The supernatant was removed for use in enzymatic assays.

2.5. Toxin accumulation and analyses

Prior to free dissolved microcystin extraction, whole mussel tissues were lyophilized and ground. All of the mussel samples were weighed before and after drying. Microcystin extraction was conducted following the method described in Vareli et al.(2012) A spiked recovery test was conducted on non-intoxicated mussel tissue. Homogeneous samples were spiked with 0.8 μg L^{-1} of microcystin-LR (Beacon Analytical Systems, Inc., Portland, ME, USA), then extracted and assessed using an enzyme-linked immunosorbent assay (ELISA) kit (Beacon Analytical Systems, Inc., Portland, ME, USA). The extraction efficiency for whole soft mussel tissue approached 90% (0.72±0.04 μg L^{-1}). The microcystin content of each sample was determined using a high-sensitivity ELISA test kit (0.1 μg L^{-1} microcystin equiv), with a detection range of 0.1-2 μg L^{-1} microcystin (Beacon Analytical Systems, Inc., Portland, ME, USA). A polyclonal antibody was used to bind microcystin and a microcystin–enzyme conjugate. The Beacon Microcystin Plate Kit is not available for the identification microcystin variants. For the ELISA, suitable volumes of each sample were dehydrated to dryness in a vacuum concentrator (SAVANT SpeedVac concentrator, SPD2010) under cold conditions, and then the residues were resuspended in 100 mL deionized water. Finally, the total microcystin contents (MC, μg g^{-1}) in each test sample were determined.

2.6. Determination of biochemical parameters

ROS levels in mussel hemocytes were detected by flow cytometry using Carboxy-H_2DCFDA (Invitrogen, C400). For each analysis, an aliquot of 400 μL hemolymph were incubated with 25 μM carboxy-H_2DCFDA in complete darkness at room temperature for 30 min, washed twice with PBS to remove the extracellular fluorescent indicators, and then re-suspended in PBS to be analyzed. Fluorescence was measured using excitation and emission (FL-1) wavelengths of 495 nm and 525 nm, respectively, in the same fluorescence reader, as described previously. ROS levels in hemocytes are expressed as arbitrary units (A.U.).

SOD activity was evaluated using a commercially available kit (kit FlukaSwitzer-land). The calibration curve was established using merchant horseradish SOD (Sigma-Aldrich, Germany). The total SOD activity was then measured at an absorbance of 450 nm.

CAT activity was detected by the dismutation of hydrogen peroxide in 0.1 M Tris buffer (pH 8.0) containing 0.5 mM EDTA and 10 mM H_2O_2 (Beutler, 1975). Results are expressed in units (U), defined as the dose of CAT per minute that resolved 1 mM of H_2O_2. The enzymatic assay was conducted for 1 min at 240 nm at 28□, and the CAT content was determined according to a standard curve.

GPx activity was estimated indirectly at 340 nm, based on the rate of NADPH oxidation by the coupled reaction of glutathione reductase (Lawrence and Burk, 1976). The mixed assay contained 600 μL buffer (50 mM potassium phosphate, 1 mM EDTA, 1 mM NaN_3, pH 7.5), 100 μL 0.2 mM decreasing glutathione (GSH), 100 μL 0.1 mM NADPH, 8 μL glutathione reductase and 20 μL sample. After incubation at 28□ for 5 min, the reaction was activated by the addition of 100 μL 0.25 mM H_2O_2. The specific activity was measured at the extinction coefficient of 6.22 mM^{-1} cm^{-1}.

Glutathione (GSH) content was measured based on the Anderson (1985) procedure, with modifications. Briefly, 10% sulfosalicylic acid (50 mL) was used to acidize the hemolymph (100 μL), followed by centrifugation at 8,000×g for 10 min. The supernatant (acid-soluble GSH) was collected, and 6 mM 5,5-dithiobis-(2-nitrobenzoic) acid (DTNB) in 0.143 M sodium sulfate buffer (pH7.5), containing 6.3 mM ethylenediaminetetraacetic acid (EDTA), was added to the supernatant. The absorbance at 412 nm was measured for 30 min at room temperature. GSH content was determined according to a standard curve, using different known concentrations of GSH.

GST activity was determined using the enzymatic method described in (Habig and Jakoby, 1981). Briefly, 1-chloro-2,4-dinitrobenzene (2 mM) and reduced glutathione (2 mM) were dissolved in potassium phosphate buffer (0.1 M, pH 7.0). The absorbance at 340 nm was monitored after a 2 min incubation at 28□ in the microplate reader. When 1 mM of the substrate is conjugated per minute, the dose of the catalyzing enzyme is defined as one unit (U) of GST activity.

LPO was quantified using the malondialdehyde (MDA) assay, based on the method adopted by Buege and Aust (1978). The thiobarbituric reactive substances (TBARS) were measured at 532 nm on a microplate reader, using malondialdehyde bis (Sigma-Aldrich, Germany) as the standard.

2.7. Statistical analyses

All statistical analyses were conducted using SPSS 22 software (SPSS Inc., Chicago, IL, USA). Data for the tested parameters (MC, ROS, MDA, GST, GSH, SOD, CAT and GPx) were recorded as the mean±standard deviation. Before statistical analyses were performed, the normality of the results was determined with a 1% risk using Shapiro-Wilk's test, and the equality of variance was determined with a 5% risk using Levine's test. If necessary, all statistical data were transformed for homoscedasticity and normality assumptions to reduce the heterogeneity of variance before analysis. The combined effects of pH and toxic algae on MC, ROS, MDA, GPx, GST, SOD, CAT and GSH levels were assessed by a three-way analysis of variance (ANOVA) to analyze intergroup differences. Multiple comparisons between averages of treatments were conducted using Tukey's honestly significant difference (HSD) to distinguish the interactions among factors. The effects of each factor were assessed individually in cases where significant interactions among factors existed. Discrimination among different treatments was conducted by principal component analysis (PCA) using XLSTAT® 2014 (Addinsoft Inc., New York, NY, USA). The significance was set at $p<0.05$.

Results

All triangle sail mussels were alive during the whole experimental period. The MC increased significantly with increasing toxic *M. aeruginosa* concentrations. At each pH level and for each sampling time point, the MC were significantly higher in mussels treated with the 100% toxic *M. aeruginosa* concentrations than those in mussels treated with the 50% toxic *M. aeruginosa* concentrations ($p<0.05$) (Fig. 1). There were no significant differences in the MC among the different pH levels in mussels treated with either the 50% or 100% toxic *M. aeruginosa* concentrations. However, after 7 d of treatment with the 100% toxic *M. aeruginosa* concentration, the MC of mussels at pH 9.0 increased significantly compared with those of mussels at pH 8.5 ($p=0.041$, $p<0.05$) and at pH 8.0 ($p=0.008$, $p<0.05$). Time showed no effects on the MC during the exposure period; however, the MC on day 21 were significantly reduced compared with those on days 7 and 14 ($p<0.05$) (Fig. 1).

The ROS levels increased gradually with increasing concentrations of toxic *M. aeruginosa* at each pH level and for each sampling time point. The highest ROS levels were obtained in mussels receiving the pH 9.0×100% toxic *M. aeruginosa* treatment (Fig. 2). The ROS levels in mussels at pH 9.0 were significantly increased compared with those in mussels at pH 8.5 and pH 8.0 for each toxic *M. aeruginosa* concentration on day 7 and 14 ($p<0.05$). However, on day 21, pH levels showed no significant effects on the ROS levels at the 0% toxic *M. aeruginosa* concentration, whereas differences were observed among the three pH levels at the 50% and 100% toxic *M. aeruginosa* concentrations. Time showed no effects on the MC during the exposure period, but ROS levels at day 21 were significantly reduced compared with those on day 7 and 14 ($p<0.05$)

(Fig. 2).

The SOD activity levels increased gradually with increasing concentrations of toxic *M. aeruginosa* at each pH level and at each sampling time point. The SOD activity levels were significantly higher at pH 9.0 than those at either pH 8.0 or 8.5 in mussels treated with the 50% toxic *M. aeruginosa* concentration on day 14. In contrast, the SOD activity levels in mussels at pH 9.0 were the lowest among the three pH levels on day 21 (p<0.05) (Fig. 3). The SOD activity levels were not significantly impacted by pH, and time showed no significant effects during the exposure period. However, after depuration, the SOD activity levels on day 21 in mussels treated with all concentrations of the toxic *M. aeruginosa* were significantly decreased compared to those on days 7 and 14 (p<0.05) (Fig. 3).

The CAT activity levels were only impacted by toxic *M. aeruginosa* treatments and showed declined significantly with increasing toxic *M. aeruginosa* concentrations. On days 7 and 14, the highest CAT activity value was observed for the pH 9.0×0% toxic *M. aeruginosa* treatment (Fig. 4). On day 7, the CAT activity levels were significantly decreased at pH 8.5 compared with those at pH 8.0 and pH 9.0 for the 0% toxic *M. aeruginosa* concentration. On day 14, the effect of pH on CAT activity levels showed opposing trends for the 0% and 50% toxic *M. aeruginosa* concentrations. CAT activity was increased significantly at pH 9.0 compared with those at pH 8.0 and pH 8.5 for the 0% toxic *M. aeruginosa* treatment, whereas CAT activity was decreased significantly at pH 9.0 compared with those at pH 8.0 and pH 8.5 for the 50% toxic *M. aeruginosa* treatment. pH did not show significant effects on CAT activity levels in the other treatment groups. Time did not show any significant effects on day 7 and 21, but on day 14. CAT activity was significantly decreased at 0% and 50% toxic *M. aeruginosa* treatment levels (p<0.05) (Fig. 4).

The GPx activity level increased gradually with increasing concentrations of toxic *M. aeruginosa* at each pH level and at each sampling time point (Fig. 5). There were no significant differences among the three pH levels at each pH level in each sampling time point. Time showed no significant effects during the exposure period; however, after depuration, GPx activity levels on day 21 in the toxic *M. aeruginosa* treatment groups decreased significantly compared with those on day 7 and 14 (p<0.05) (Fig. 5).

The content of GSH increased gradually with increasing concentrations of toxic *M. aeruginosa* and over time. The highest GSH levels were obtained on day 21 in the 100% toxic *M. aeruginosa* groups, and the lowest GSH levels were obtained on day 7 in the 0% toxic *M. aeruginosa* groups, regardless of pH (Fig. 6). There were no significant differences observed in GSH levels among the three pH levels for each toxic *M. aeruginosa* concentration and at each sampling time point. However, on day 21, GSH contents in the 100% toxic *M. aeruginosa* treatment groups were reduced at pH 9 compared with those at pH 8 (p<0.05) (Fig. 6).

The GST activity levels were only influenced by the concentration of toxic *M. aeruginosa*. With increasing concentrations of the toxic algae, the GST activity levels increased for each pH treatment and at each sampling time point (p<0.05) (Fig. 7). The highest values were observed for the 100% toxic *M. aeruginosa* treatments.

The MDA levels increased gradually with increasing concentrations of toxic *M. aeruginosa* at each pH level during the exposure period. However, on day 21, there were no significant differences between the 0% and 50% toxic *M. aeruginosa* concentrations at either pH 8.0 or pH 8.5. During the exposure period, for the 0% and 50% toxic *M. aeruginosa* concentrations, the MDA levels at pH 9.0 were significantly increased compared with those at pH 8.0 and pH 8.5

(p<0.05) (Fig. 8). There were no significant differences observed in the MDA levels on days 7 and 14 during the exposure period; however, the MDA levels on day 21 were significantly lower than those on day 7 and 14 (p<0.05) (Fig. 8).

According to the three-way ANOVA results, every parameter could be significantly affected by toxic *M. aeruginosa*. Time showed no significant effects on GST activity. pH levels only impacted the ROS and MDA levels significantly (Table 1). Significant interactions were observed for both time and toxic *M. aeruginosa* treatments for most of the tested parameters, with the exception of GST activity. The interaction between pH levels and toxic *M. aeruginosa* treatments had extremely significant effects on the ROS and MDA levels, as well as CAT activity levels (p<0.001). In contrast, no significant differences were observed for the GSH, GST, and GPx activity levels or for MC. The interaction between time and pH values also had no significant effects on the GST, CAT and GPx activity levels or on the MC. Statistically significant interactions among time course, pH and toxic *M. aeruginosa* treatments were observed to affect the ROS, MDA and GSH levels, as well as the CAT activity levels (p<0.05) (Table 1).

The PCA performed for all of the tested parameters showed that two principal components explained 96.48% of the total variance (Fig. 9). PC1was responsible for 83.59% of the total variance and was positively influenced by all of the parameters except CAT activity, which can be observed on the left side of Fig. 9. During the entire exposure process, PC2 was only responsible for 12.89% of the total variance and was positively correlated with GSH levels and GST activities.

According to the three-way ANOVA and PCA results, during the exposure period (7 and 14 d of exposure), the A (pH 8.0×0% toxic *M. aeruginosa*), B (pH 8.5×0% toxic *M. aeruginosa*), and C (pH 9.0×0% toxic *M. aeruginosa*) treatments were found in the same group. Even after the 7-day depuration period, the A21, B21, and C21 treatments were similar to the exposure period treatments (A7, B7, C7, A14, B14, and C14). When the concentration of toxic *M. aeruginosa* was increased to 50%, after 7 d of exposure, D7 (pH 8.0×50% toxic *M. aeruginosa*, 7 d exposure), E7 (pH 8.5×50% toxic *M. aeruginosa*, 7 d exposure) and F7 (pH 9.0×50% toxic *M. aeruginosa*, 7 d exposure) belonged to the same group, which was similar to the group consisting of D14 (pH 8.0×50% toxic *M. aeruginosa*, 14 d exposure), E14 (pH 8.5×50% toxic *M. aeruginosa*, 14 d exposure) and F14 (pH 9.0×50% toxic *M. aeruginosa*, 14 d exposure) but was significantly different compared with the control treatment and A7(pH 8.0×0% toxic *M. aeruginosa*,7 d exposure). At the same concentration of toxic *M. aeruginosa*as above, after the 7-day depuration period, D21 (pH 8.0×50% toxic *M. aeruginosa*, 7 d depuration), E21 (pH 8.5×50% toxic *M. aeruginosa*, 7 d depuration) and F21 (pH 9.0×50% toxic *M. aeruginosa*, 7 d depuration) form a group separate from the exposure groups (the group containing D7, E7 and F7 and the group containing D14, E14 and F14). The same situation applied to the 100% toxic *M. aeruginosa* exposure.

Discussion

Water eutrophication is a complex phenomenon able to affect frequency and intensity of harmful algal blooms (HAB) altering several environmental factors, such as pH, oxygen values, and so on. Consequently, it is necessary to comprehend the interaction of these factors, which may bring in either antagonistic or synergistic effects of aquatic organisms (Sui et al., 2017). The combined influences of two principal stressors, harmful algae *M. aeruginosa* and high pH levels, on the ecophysiology of the freshwater *H. cumingii* triangle sail mussel were reported for the first time in the present study. High pH did not influence total microcystin accumulation in mussels, since MC was not different in mussels under high pH + *M. aeruginosa* condition when comparing to *M. aeruginosa* alone exposed mussels. Braga et al. (2018) found lower algal toxin levels in mussels *Mytilus galloprovincialis* under low pH. The authors attributed lower accumulation in reduced pH to alterations in the biological mechanisms of uptake or elimination. On the contrary, our results suggest that the accumulation of microcystins was not affected by high pH, perhaps because the uptake and removal mechanisms were not affected by high pH.

ROS is a biomarker of oxidative stress, which can result in the loss of the mitochondrial membrane potential, calpain and Ca^{2+}/calmodulin-dependent protein kinase II transition and activation, and apoptosis. In this study, both high pH and *M. aeruginosa* could induce high content of ROS. This is similar to an increase in ROS levels observed in *Elliptio complanate* hemocytes exposed to crude extracts of *M. aeruginosa* (Gelinas et al., 2014) and an increase in *Crassostrea gigas* (Cao et al., 2018a) and *Mytilus coruscus* exposed to shift pH (Wu et al., 2018). Moreover, significant interactions were found between pH and *M. aeruginosa*, which indicated that under high pH conditions, toxic *M. aeruginosa* could induce *H. cumingii* to produce more ROS. Under varied pH, that hazardous material could stimulate shellfish to generate more ROS were also reported by Cao et al. (2018b), they suppose the synergetic effects of shift pH and hazardous material would bring a more serious threat to the immune system.

The antioxidant defense response occurs simultaneously with the generation of oxygen radicals. Antioxidant enzymes (such as SOD, CAT and GPx) and antioxidants (such as GSH) constitute the antioxidant defense system. SOD, which is the first and most important line of defense in the antioxidant system, catalyzes the dismutation of O^{2-} into H_2O_2 and water, stimulating CAT and GPx activities. CAT and GPx catalyze the conversion of H_2O_2 into O_2 and water. GSH is the most abundant, cytosolic, low-molecular weight scavenger involved in the antioxidant defense system, which acts on ROS neutralization directly or serves as a cofactor of GSH-dependent enzymes. In this study, accompany with cellular ROS release, SOD and GPx activities upregulate, but CAT activities downregulate. These results are consistent with Hu et al. (2015), which indicated that toxic *M. aeruginosa* induced antioxidant reaction by up-regulating SOD and GPx activities, whereas the down-regulation of CAT activity may be contributed to enzyme inactivation caused by microcystins or high concentrations of contaminants in tissue (Pinho et al., 2005).The effects of pH shift on shellfish oxidative stress actions was reported in abundant papers and results varied among different species (Liao et al., 2018; Velez et al., 2016). In present study, the mussels did not show significant changes in antioxidant enzymes activity between tested conditions, despite a non-significant increase in high pH conditions compared to control. Similar results were found in oysters *C. angulate* (Moreira et al., 2016). However, other studies indicated an upregulated proteomic response in oysters *C. virginica* oysters to shift pH in antioxidant enzymes (Tomanek, 2011; Tomanek et al., 2015). In *Scrobicularia plana*, Freitas et al. (2015) found antioxidant enzymes SOD activity decreased significantly under shift pH, whereas a significant increase in CAT and GPx activity. Additionally, significant interaction between pH and *M. aeruginosa* was only found in CAT, this implied that the effect of *M. aeruginosa* on CAT activity may depend on pH.

MDA is the final product of lipid peroxidation and an important indicator of oxidative damage of cell membrane. In the present study, MDA content was significantly affected by pH, toxic *M. aeruginosa* and the interaction between them. The phenomenon implied that under high pH and *M. aeruginosa* stress, although mussel *H. Cumingii* triggered antioxidant response, they could not avoid oxidative damages.

Glutathione S-transferases (GSTs), which belong to multifunctional protein families, play key roles in the detoxification reactions to harmful xenobiotic and endobiotic compounds in organisms. Our study showed that GST activities were only significantly affected by *M. aeruginosa*, not high pH and the interaction between them. The result agreed with Hu et al. (2015) in whose research GST activity was closely correlated with the concentration of *M. aeruginosa*, not closely correlated with DO concentrations. On the other hand, Han et al. (2018) found significantly elevated GST activities in the white shrimp *Litopenaeus vannamei* exposed to low or high pH environment, and explained GST facilitated the elimination of products of oxidative damage induced by shift pH. These researches imply that each taxon may respond differently to shift pH in terms of GST.

These PCA results suggest that pH value did not have significant effects on *H. cumingii* during treatment with 0% toxic *M. aeruginosa*. But toxic *M. aeruginosa* could affect *H. cumingii* in the exposure stage and had a profound influence on the process of mussel acclimatization to normal conditions. The influence of toxic *M. aeruginosa* was more apparent when compared with the control group, treated with 0% toxic *M. aeruginosa* (exposure and depuration days).

Conclusion

Astonishingly, during the entire experimental period, no mussels died, suggesting that *H. cumingii* have a high tolerance to both high pH values (9.0) and the presence of toxic *M. aeruginosa*. ROS and MDA levels, which are markers of oxidative damage, could be affected not only by toxic *M. aeruginosa* but also by pH value, suggesting that both toxic *M. aeruginosa* and high pH values could cause oxidative stress in *H. cumingii*. GST, the most important microcystin detoxifying enzyme, was only significantly affected by toxic *M. aeruginosa*. but not pH. The enzymatic activation of GST, SOD and GPx, as well as the content of GSH, which are involved in immune mechanisms, may be potential influencing factors for microcystin. To reduce the damage of ROS from microcystin, O_2 are dismutated by SOD to H_2O_2 which is reduced to water and molecular oxygen by CAT or is neutralized by GPx that catalyses the reduction of H_2O_2 to water and organic peroxide to alcohols using glutathione (GSH) as a source of reducing equivalent. This experiment explored the effects of toxic *M. aeruginosa* on aquatic shellfish,and found short term ingestion of toxic *M. aeruginosa* is not fatal to the triangle sail mussel *H. cumingii*, mussel also could recover to heath. This brings us new ideas for river and lake management, suggesting that we could use *H. cumingii* to repair the polluted rivers and lakes in the future. Unfortunately, this is a short-term study that focuses only on the detoxification and antioxidant capacity of shellfish. In the future, we will further study the effects of *M. aeruginosa* on shellfish feeding, reproduction and growth in order to clarify the impact of eutrophication on Lake ecosystem.

Declaration of interest

None

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Table 1

Summary of three-way ANOVA results for effects of pH, toxic *M. aeruginosa* treatments (M) and sampling time points (T) on total Microcystin (MC), Reactive oxygen species (ROS), Superoxide dismutase (SOD), Catalase (CAT), Glutathione peroxidase (GPX), Glutathione (GSH), Gultathione-S-transferases (GST), Malondialdehyde (MDA), as a function of time. pH: 8.0, 8.5 and 9.0; toxic (M): 0%, 50% and 100%; sampling time points: Day 7, Day 14 and Day 21.

Factor	Degrees of freedom	Mean square	F	P
MC				
T	2	1208.767	733.041	<0.001
M	2	15917.899	9653.205	<0.001
pH	2	4.437	2.691	0.077
T×M	4	906.175	549.538	<0.001
T×pH	4	2.866	1.738	0.155
M×pH	4	2.078	1.26	0.297
T×M×pH	8	1.859	1.128	0.36
ROS				
T	2	1.492	184.701	<0.001
M	2	18.645	2307.697	<0.001
pH	2	1.185	146.708	<0.001
T×M	4	0.157	19.47	<0.001
T×pH	4	0.022	2.677	0.041
M×pH	4	0.059	7.252	<0.001
T×M×pH	8	0.028	3.428	0.003
SOD				
T	2	15351.856	1187.858	<0.001
M	2	95846.968	7416.212	<0.001
pH	2	24.124	1.867	0.164
T×M	4	4102.077	317.401	<0.001
T×pH	4	39.648	3.068	0.024
M×pH	4	6.019	0.466	0.761
T×M×pH	8	22.265	1.723	0.114
CAT				
T	2	27.102	62.763	<0.001
M	2	438.357	1015.165	<0.001
pH	2	0.682	1.58	0.215
T×M	4	8.881	20.566	<0.001
T×pH	4	0.429	0.993	0.419
M×pH	4	3.347	7.751	<0.001
T×M×pH	8	1.121	2.596	0.018

GPX				
T	2	387.992	127.619	<0.001
M	2	2576.203	847.366	<0.001
pH	2	1.03	0.339	0.714
T×M	4	117.286	38.578	<0.001
T×pH	4	0.793	0.261	0.902
M×pH	4	3.68	1.21	0.317
T×M×pH	8	2.981	0.98	0.461
GSH				
T	2	42.448	1447.657	<0.001
M	2	117.386	4003.384	<0.001
pH	2	0.011	0.391	0.678
T×M	4	7.78	265.324	<0.001
T×pH	4	0.105	3.585	0.012
M×pH	4	0.073	2.478	0.055
T×M×pH	8	0.077	2.643	0.016
GST				
T	2	0.003	0.057	0.944
M	2	243.389	4787.178	<0.001
pH	2	0.035	0.695	0.503
T×M	4	0.048	0.95	0.443
T×pH	4	0.075	1.467	0.225
M×pH	4	0.016	0.307	0.872
T×M×pH	8	0.091	1.78	0.101
MDA				
T	2	0.672	32.329	<0.001
M	2	29.427	1416.146	<0.001
pH	2	91.557	4406.059	<0.001
T×M	4	0.092	4.411	0.004
T×pH	4	0.142	6.813	<0.001
M×pH	4	8.905	428.538	<0.001
T×M×pH	8	0.059	2.828	0.011

Figure legends

Fig. 1. Total microcystin content (μg gDW^{-1}) of *H. cumingii* soft tissue in different treatments for 14 d exposure (sampling time points: Day 7 and Day 14) and 7 d depuration duration (sampling time point: Day 21).

Fig. 2. ROS level of *H. cumingii* hemocytes in different treatments for 14 d exposure (sampling time points: Day 7 and Day 14) and 7 d depuration duration (sampling time point: Day 21).

Fig. 3-8. Variations of physiological parameters (SOD, CAT, GPX, GSH, GST and MDA) in *H. cumingii* in different treatments for 14 d exposure (sampling time points: Day 7 and Day 14) and 7 d depuration duration (sampling time point: Day 21).

Fig. 9. Results of the PCA for the two principal components produced by tested parameters variables in mussels of nine treatments for different sampling time points. (a) Plot of variable vectors, (b) plot of scores of different treatments: pH 8.0 × 0% toxic *M. aeruginosa* (A), pH 8.5 × 0% toxic *M. aeruginosa* (B), pH 9.0 × 0% toxic *M. aeruginosa* (C), pH 8.0 × 50% toxic *M. aeruginosa* (D), pH 8.5 × 50% toxic *M. aeruginosa* (E), pH 9.0 × 50% toxic *M. aeruginosa* (F), pH 8.0 × 100% toxic *M. aeruginosa* (G), pH 8.5 × 100% toxic *M. aeruginosa* (H), pH 9.0 × 100% toxic *M. aeruginosa* (I).

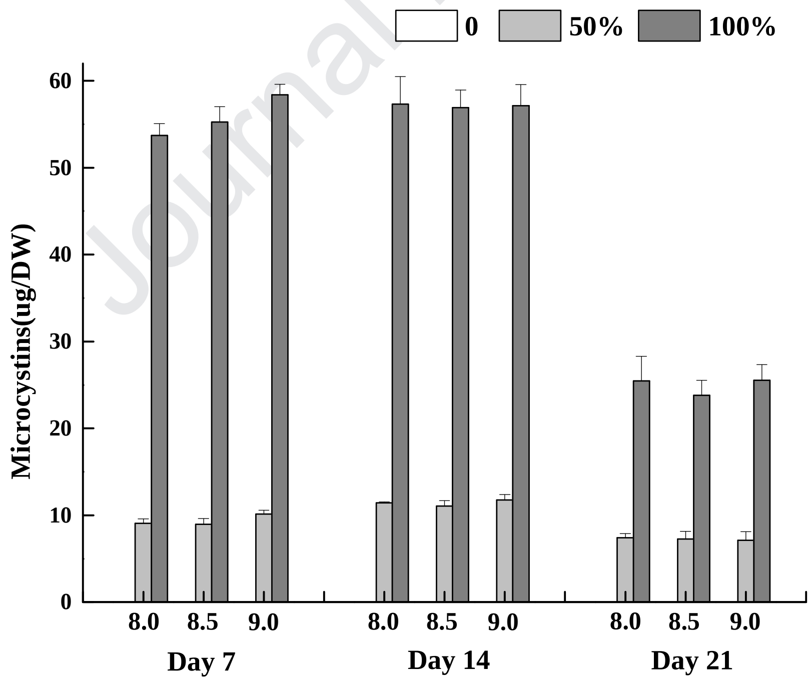
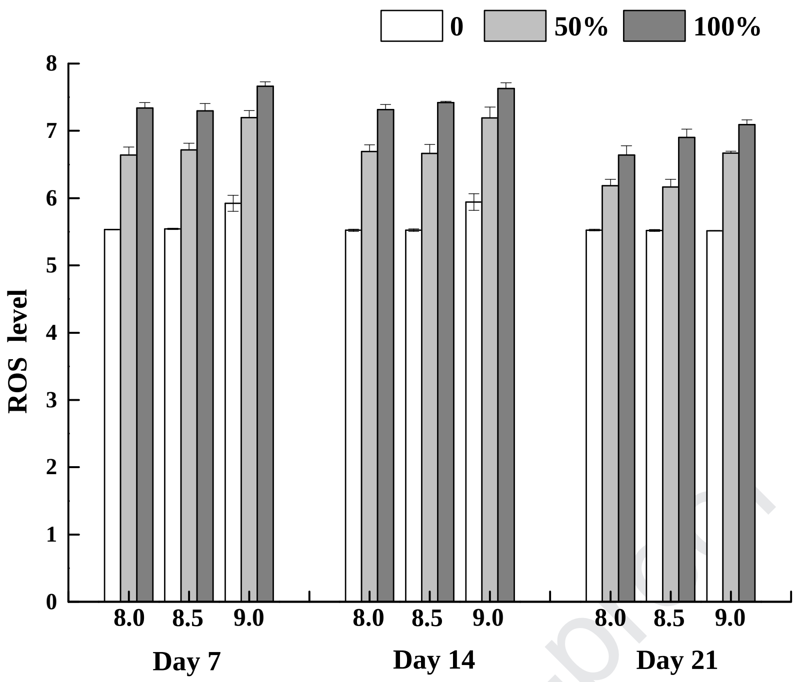
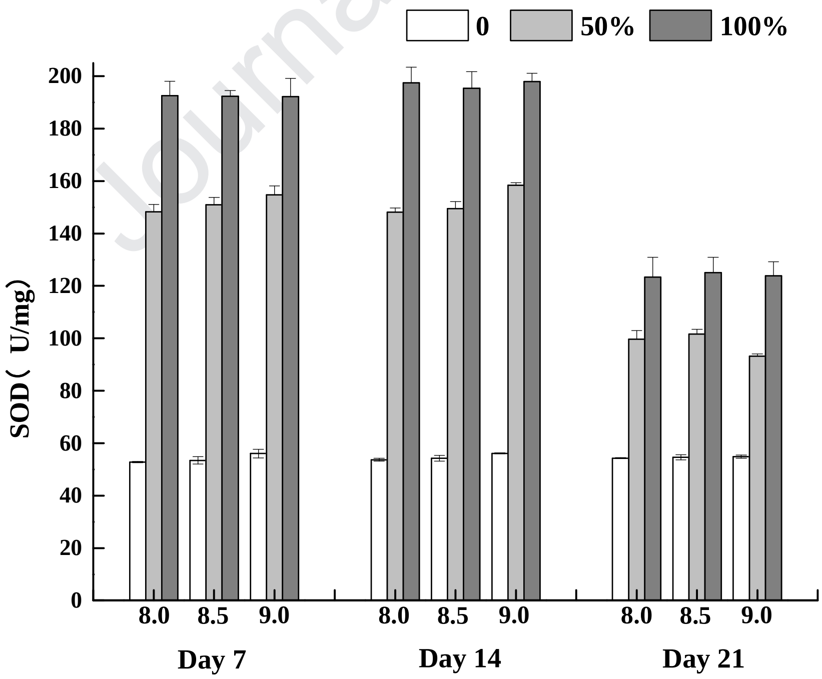


Fig. 1.



0
50%
100%
ROS level
8
7
6
5
4
3
2
1
0
8.0
8.5
9.0
Day 7
Day 14
Day 21

Fig. 2.



0
50%
100%
SOD（U/mg）
200
180
160
140
120
100
80
60
40
20
0
8.0
8.5
9.0
Day 7
Day 14
Day 21

Fig. 3.

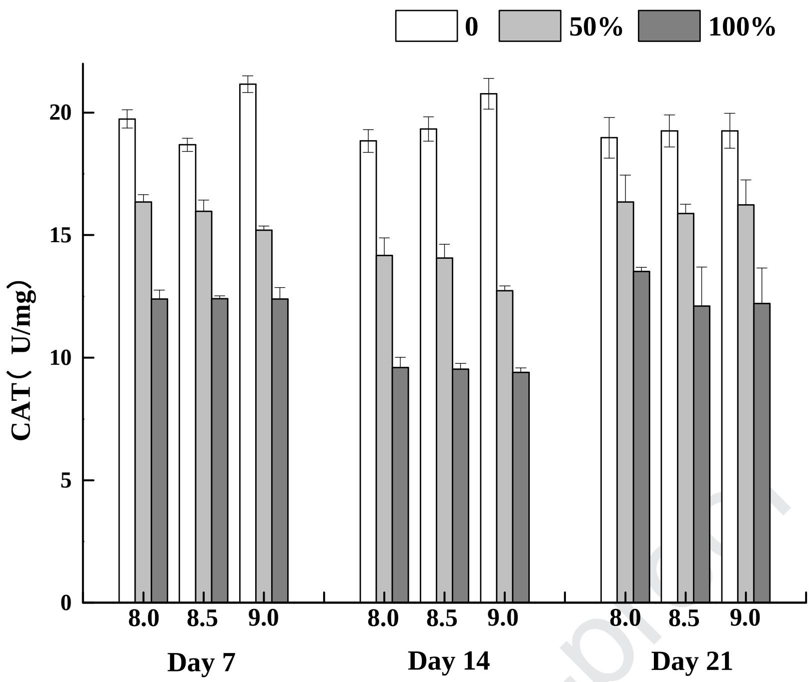

0
50%
100%
CAT（ U/mg)
0
5
10
15
20
8.0
8.5
9.0
8.0
8.5
9.0
8.0
8.5
9.0
Day 7
Day 14
Day 21

Fig. 4.

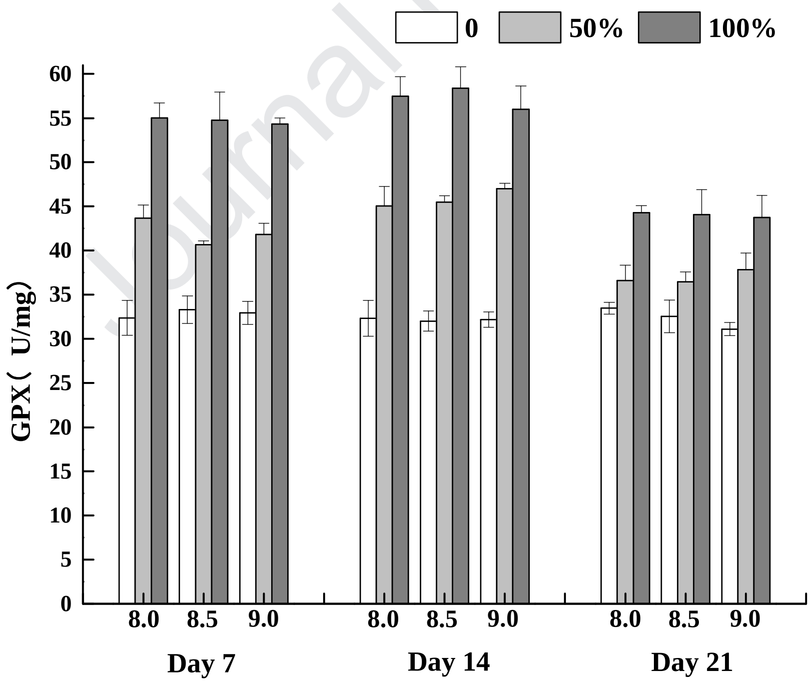

0
50%
100%
GPX（ U/mg)
0
5
10
15
20
25
30
35
40
45
50
55
60
8.0
8.5
9.0
8.0
8.5
9.0
8.0
8.5
9.0
Day 7
Day 14
Day 21

Fig. 5.

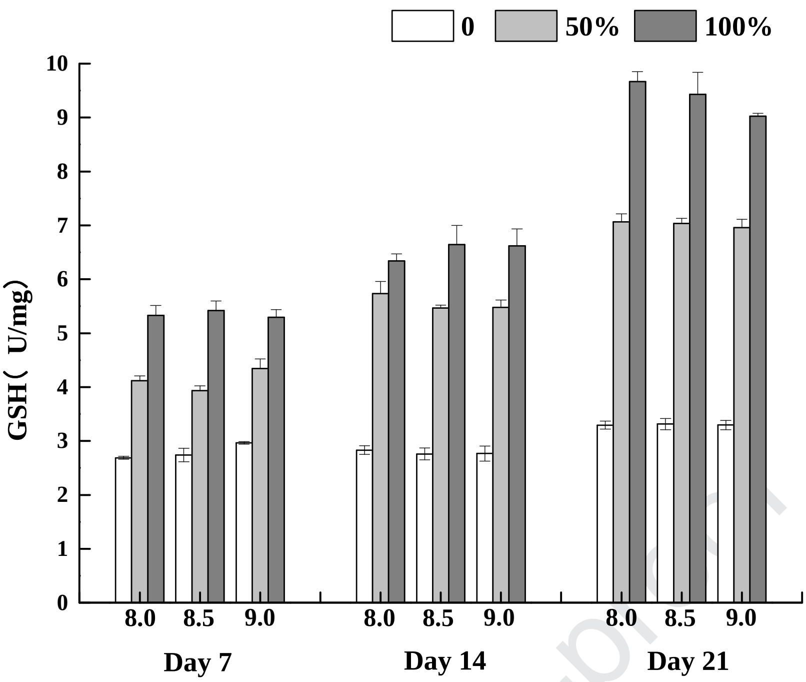

0
50%
100%
GSH（ U/mg）
10
9
8
7
6
5
4
3
2
1
0
8.0
8.5
9.0
Day 7
Day 14
Day 21

Fig. 6.

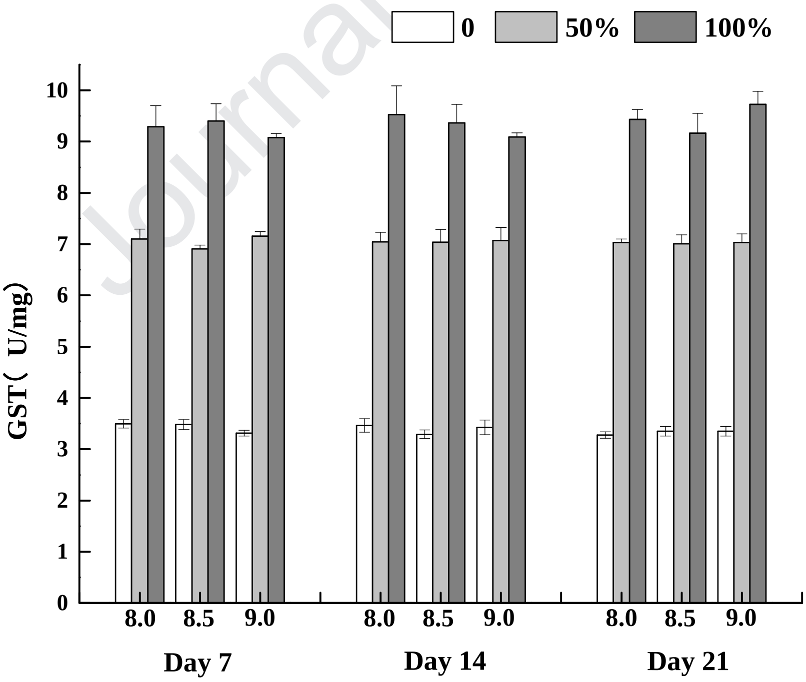
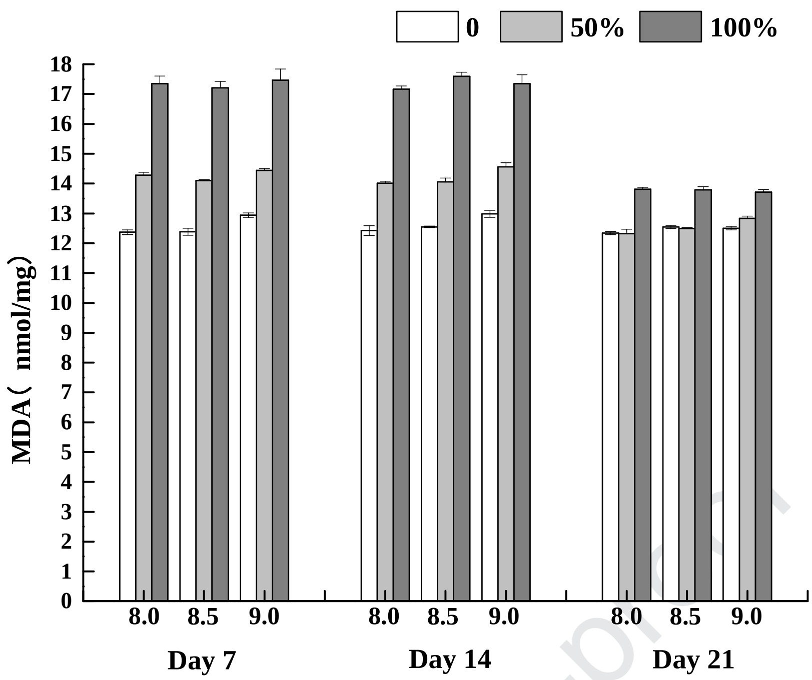
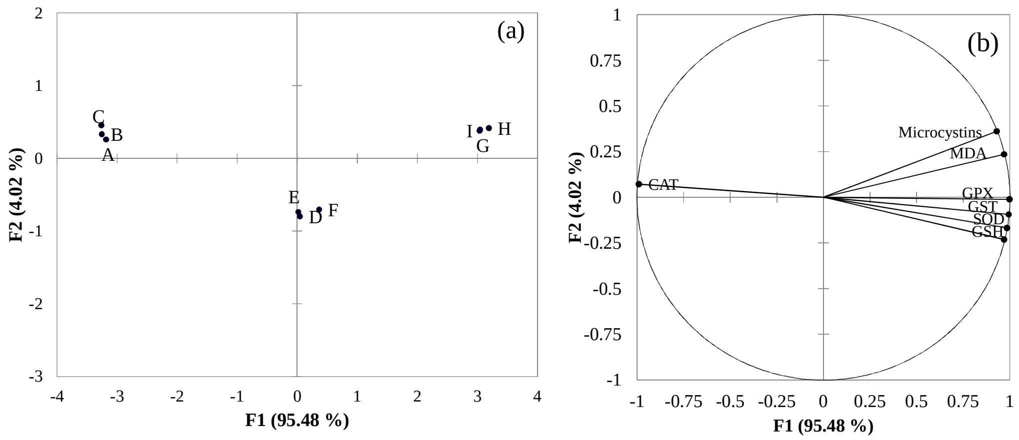

0
50%
100%
GST（ U/mg）
10
9
8
7
6
5
4
3
2
1
0
8.0
8.5
9.0
Day 7
Day 14
Day 21

Fig. 7.



0
50%
100%
MDA（nmol/mg)
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
8.0
8.5
9.0
Day 7
Day 14
Day 21

Fig. 8.



(a)
F2 (4.02 %)
F1 (95.48 %)
A
B
C
D
E
F
G
H
I
(b)
Microcystins
MDA
GPX
GST
SOD
GSH
CAT
F2 (4.02 %)
F1 (95.48 %)

Fig. 9.

HIGHLIGHTS

The comprehensive effects of toxic cyanobacteria and high pH on mussels were assessed.

Interaction between cyanobacteria and high pH on physiological indicator were found.

Compare to high pH, toxic *M. aeruginosa* induce more severe oxidative stress response.

Toxic algae or high pH exposure history showed latent effects on *Hyriopsis cumingii*.

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Antioxidant responses of triangle sail mussel *Hyriopsis cumingii* exposed to harmful algae *Microcystis aeruginosa* and high pH”